

Microcontroller PIC16F72 is the heart of the switch. It is an 8-bit, low-cost, high-performance, Flash microcontroller. Its key features are 2 kB of Flash program memory, 128 bytes of RAM, eight interrupts, three input/output (I/O) ports, three timers and a five-channel 8-bit analogue-to-digital converter (ADC). There are 22 I/O pins, which are user-configurable for input/output on pin-to-pin basis. Architecture is RISC, and there are only 35 powerful instructions.

System clock plays a significant role in operation of the microcontroller. A 4MHz quartz crystal connected between pins 9 and 10 provides the basic clock to the microcontroller (IC1).

Two 7-segment displays (DIS1 and DIS2) are used to display the time in minutes. Port pins RB2, RB3, RA0, RA1, RA2, RB1 and RB0 are connected to segment pins 'a' through 'g' of display DIS1, respectively. Ports pin RC6, RC7, RC1, RC2, RC3, RC5 and RC4 are connected to segment pins 'a' through 'g' of display DIS2, respectively.

PARTS LIST

Semiconductors:

IC1	- PIC16F72 microcontroller
IC2	- 7805, 5V regulator
T1	- BC547 npn transistor
D1-D5	- 1N4007 rectifier diode
DIS1, DIS2	- LTS542 common-anode 7-segment display
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R16	- 10-kilo-ohm
R2-R15	- 220-ohm
R17	- 4.7-kilo-ohm
R18, R19	- 330-ohm

Capacitors:

C1	- 10 μ F, 16V electrolytic
C2, C3	- 22pF ceramic disk
C4	- 1000 μ F, 35V electrolytic
C5	- 0.1 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
RL1	- 12V, single-changeover relay
X _{cr}	- 4MHz crystal
S1-S5	- Push-to-on tactile switch

Switches S2 (start/stop), S3 (select), S4 (decrement) and S5 (increment) are connected to port pins RB4 through RB7 of the microcontroller, respectively. Port pin RC0 of the microcontroller is used to control relay RL1 with the help of transistor T1. When port pin RC0 is

high, transistor T1 drives into saturation and 12V-relay RL1 energises to connect the load to power supply. Diode D5 acts as a free-wheeling diode.

To derive the power supply for the circuit, the 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C4 and regulated by IC 7805 (IC2). Capacitor C5 is used to bypass the ripples present in the regulated supply. LED2 gives power-on indication. Resistor R19 limits the current through LED2. Switch S1 is used for manual reset.

Set the time using switch S4 for decrement and switch S5 for increment. The time is indicated on 7-segment displays DIS1 and DIS2. To start timing count-down, press start/stop switch S2. Relay RL1 energises to switch on the appliance and LED1 glows. If you press start/stop switch S2 again, the count-down process will

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